Real-World Validation: 500-Question Benchmark

N-ATLaS Health QA against base N-ATLaS on held-out AfriMed-QA questions.

1. Summary

Base N-ATLaS and the fine-tuned model were scored on the same 500 held-out African medical multiple-choice questions. The validation rule for Problem Statement 3 is a benchmark of 500 or more examples against base N-ATLaS.

Model	Correct	Accuracy	95% confidence interval (Wilson)
Base N-ATLaS (NCAIR1/N-ATLaS)	252 / 500	50.4%	46.0% to 54.8%
N-ATLaS + this adapter	303 / 500	60.6%	56.2% to 64.8%
Chance (random guessing)		21.2%	

The fine-tuned model was more accurate than base N-ATLaS by 10.2 percentage points on the same 500 questions. The exact McNemar p-value is 1.4 x 10^-6, so the difference is statistically significant at the 0.05 level.

2. The benchmark

- Source: AfriMed-QA v2 (intronhealth/afrimedqa_v2), a pan-African medical question dataset. Licence CC BY-SA 4.0. Olatunji et al., 2024, arXiv:2411.15640.
- Questions used: multiple-choice questions with exactly one correct answer. Of 4039 multiple-choice rows, 3722 were usable. 316 were dropped because they have more than one correct answer or an answer key that does not match an option, and 1 because its options were empty.
- Split: 500 questions held out at random (seed 42) from all usable single-answer questions before training; the dataset's official splits were not used because they hold 124 usable train and 3598 usable test questions, so the remaining questions, most of them from the official test split, were used for training.
- Leakage check: questions were de-duplicated on normalised text before the split, and the script stops with an error if any benchmark question also appears in the training set. It did not fire.
- The IDs of the 500 questions are published in benchmark_500_ids.csv, and both models' answers to every question are in benchmark_500_predictions.csv.

3. How each model was scored

- Both models got the identical prompt for each question, built with N-ATLaS's own chat template.
- The predicted answer is the option letter with the highest next-token probability, restricted to the letters that exist for that question. One forward pass per question, with no sampling, so the scores are deterministic. The base score was reproduced exactly on a second run.
- Both models used the same 4-bit NF4 quantised weights.
- Chance accuracy is 21.2%: the average of one divided by the number of options.

4. Statistics

- Confidence intervals are 95% Wilson score intervals.
- The two models are compared with an exact McNemar test, which uses only the questions where they disagree. This is the correct test for two models scored on the same questions.
- Base right and fine-tuned wrong: 30 questions. Fine-tuned right and base wrong: 81 questions. Exact two-sided p-value: 1.4 x 10^-6.

5. Accuracy by medical specialty

Specialties with at least 10 benchmark questions. These groups are small, so individual rows are noisy and should not be read as significant on their own.

Specialty	Questions	Base N-ATLaS	Fine-tuned	Change (points)
Obstetrics and Gynecology	71	36.6%	54.9%	+18.3
General Surgery	65	40.0%	55.4%	+15.4
Pediatrics	59	42.4%	55.9%	+13.5
Pathology	33	57.6%	66.7%	+9.1
Cardiology	33	57.6%	54.5%	-3.1
Infectious Disease	32	62.5%	59.4%	-3.1
Psychiatry	26	69.2%	65.4%	-3.8
Neurology	24	58.3%	75.0%	+16.7
Endocrinology	17	47.1%	41.2%	-5.9
Gastroenterology	17	82.4%	70.6%	-11.8
Pulmonary Medicine	16	50.0%	68.8%	+18.8
Nephrology	13	61.5%	76.9%	+15.4
Hematology	11	27.3%	45.5%	+18.2
Internal Medicine	10	80.0%	90.0%	+10.0

6. Limits

- This is a research model. It is not a medical device and must not be used for diagnosis or treatment.
- Training and testing were in English only. Nothing was evaluated in Yoruba, Hausa or Igbo, so no claim is made about those languages.
- Only multiple-choice questions with a single correct answer were used. Questions with more than one correct answer were excluded. Open-ended and consumer questions were not tested.
- The model is scored by which option letter it rates most likely. This measures exam-style knowledge, not the quality or safety of free-text medical advice.
- The training data includes questions from the official AfriMed-QA test split, so this adapter must not be compared with published AfriMed-QA results. The only fair comparison is on the 500 held-out questions.
- One training run, one seed (42), 1 epoch. No hyperparameter search was done, and nothing was tuned on the 500 benchmark questions.
- Both models were scored with 4-bit quantised weights. Full-precision scores may differ slightly.
- The 500 questions come from the same dataset as the training questions. Exact duplicates were removed, but near-duplicate or closely related questions were not checked for.

7. Reproducing the numbers

Run natlas_pipeline.py (or the notebook natlas_health_kaggle.ipynb) from https://github.com/Emart29/emart-health with a Hugging Face token that has access to N-ATLaS and AfriMed-QA. The seed, the split and the scoring are fixed, so the base score reproduces exactly. Training on a different GPU or library version may shift the fine-tuned score slightly.